

Cell: The Building Block of Life

Test Objective

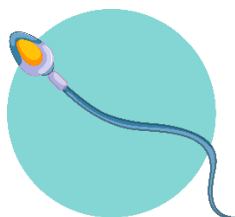
Time : 30 minutes

Maximum marks : 20

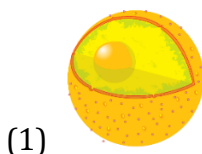
Instructions

- This test contains 20 questions.
- Each question has **FOUR** options (1), (2), (3) and (4). **ONLY ONE** of these four options are correct.
- For each question, marks will be awarded in one of the following categories:
Full Marks : +1 : If only correct answer is given.
Zero Marks : 0 : If no answer is given.
Negative Marks : There is no negative marking.

- _____ is the basic structural and functional unit of living beings. choose the correct one from the following given options.
 (1) Cell (2) Organ
 (3) Cell organelle (4) All of the above
- Which of the given options represents the longest cell in the human body?
 (1) Bone cell (2) Muscle fibre
 (3) Epithelial cell (4) Neuron
- "All cells arise from pre – existing cells" was stated by –
 (1) Robert Hooke (2) Robert brown (3) Huxley (4) Rudolf Virchow
- Identify the following diagram.



- Cell organelle which plays a role in formation of acrosome of sperm is –



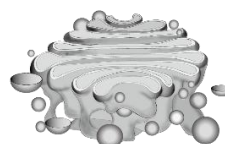
(1)



(3)

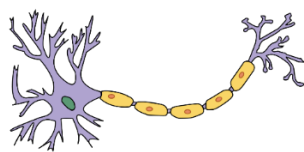


(2)



(4)

6. Given diagram is for –



(1) bone cell

(2) nerve cell

(3) blood cell

(4) fat cell

7. The largest cell is –

(1) ostrich egg

(2) brain cell

(3) hemp fibres

(4) none of these

8. Protoplasm is composed of –

(1) cytoplasm and cell membrane

(2) cytoplasm and nucleus

(3) all the cell organelles

(4) cytoplasm and cell wall

9. When a cell is placed in hypertonic solution it shrinks because –

(1) salt solution enters the cell.

(2) cytoplasm of the cell begins to decompose.

(3) water comes out of the cell.

(4) all of these

10. Identify an animal cell which is responsible for carry messages from one part of body to another and have long extensions for this purpose.

(1) WBC

(2) Nerve cell

(3) RBC

(4) Egg cell

11. Structure of cell membrane explained by –

(1) Robert brown

(2) Singer and Nicolson

(3) Purkinje

(4) Golgi

12. Given diagram helps in synthesis of food in plants. identify this –



(1) mitochondria

(2) chloroplast

(3) nucleus

(4) vacuole

13. Shrinking of protoplasm in a plant cell is referred to as –

(1) plasmolysis

(2) endosmosis

(3) diffusion

(4) none of these

14. Cell wall is present in all except –
 (1) plant cell (2) fungi
 (3) animal cell (4) blue green algae
15. Double membrane is absent in –
 (1) mitochondria (2) chloroplast
 (3) nucleus (4) lysosome
16. Match the column – I with column – II and choose the correct option.

Column – I		Column – II	
(a)	Largest cell	(p)	Mycoplasma
(b)	Smallest cell	(q)	Protoplasm
(c)	Robert Hooke	(r)	Ostrich egg
(d)	J.E. Purkinje	(s)	Micrographia

- (1) (a) – (r), (b) – (p), (c) – (s), (d) – (q)
 (2) (a) – (r), (b) – (q), (c) – (s), (d) – (p)
 (3) (a) – (s), (b) – (p), (c) – (r), (d) – (q)
 (4) (a) – (r), (b) – (s), (c) – (p), (d) – (q)
17. Robert Hooke –
 (1) observed honeycomb like structures in cork.
 (2) wrote a book micrographia.
 (3) discovered nucleus.
 (4) both (1) and (2)
18. _____ was the first one to observe free living cells. the term cell was given by _____. Cell theory was proposed by _____ and _____. All cells arise from pre-existing cells was suggested by _____.
- select the correct sequence of names to complete the above paragraph.
 (1) Robert Hooke, Virchow, Anton van Leeuwenhoek, Schleiden, Schwann
 (2) Anton van Leeuwenhoek, Robert Hooke, Schleiden, Schwann, Virchow
 (3) Robert Hooke, Virchow, Schleiden, Schwann, Anton van Leeuwenhoek
 (4) Anton van Leeuwenhoek, Virchow, Schleiden, Schwann, Robert Hooke
19. The cork cells as seen by Robert Hooke were published in his book –
 (1) origin of species (2) plant kingdom
 (3) genera plantarum (4) micrographia
20. The cell theory states that –
 (1) every new cell arises from pre-existing cells.
 (2) bodies of living beings are made up of cells.
 (3) activities of an organism are the sum total of activities of its cells.
 (4) all of the above

Test Subjective**Time : 20 minutes****Maximum marks : 10****Instructions**

- This test contains 6 questions.
- Q.1 to Q.3 are one-mark questions, to be answer in about one word or one sentence.
- Q.4 & Q.5 are two-mark questions, to be answer in about 50 words.
- Q.6 is three-mark question, to be answer in about 80 words.

1. Name two cell organelles which are covered by double membrane.
2. Name two examples of unicellular organisms.
3. When a perfume bottle in a room, it spreads uniformly in the room. Which process responsible for this?
4. Write one function of the following plastids –
(i) Leucoplasts
(ii) Chloroplasts
5. Why mitochondria is called a semi-autonomous organelle?
6. What is active transport and passive transport?

Test Objective Solutions

Answer Key

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	4	4	4	2	1	2	3	2	2	2	1	3	4
Question	16	17	18	19	20										
Answer	1	4	2	4	4										

1. Option (1)

Cell is the basic structural and functional unit of living beings.

2. Option (4)

Neuron is the longest cell in the human body.

3. Option (4)

“All cells arise from pre-existing cells” was stated by Rudolf Virchow.

4. Option (4)

It is sperm cell.

5. Option (4)

Golgi complex plays a role in formation of acrosome of sperm.

6. Option (2)

Given diagram is nerve cell.

7. Option (1)

The largest cell is ostrich egg.

8. Option (2)

Protoplasm is composed of cytoplasm and nucleus.

9. Option (3)

When a cell is placed in hypertonic solution it shrinks because water comes out of the cell due to osmosis process.

10. Option (2)

Nerve cell is responsible for carry messages from one part of body to another and have long extensions for this purpose.

11. Option (2)

Structure of cell membrane explained by singer and Nicolson.

12. Option (2)

chloroplast helps in synthesis of food in plants.

13. Option (1)

Shrinking of protoplasm in a plant cell is plasmolysis.

14. Option (3)

Cell wall is absent in animal cell.

15. Option (4)

Double membrane is absent in lysosome.

16. Option (1)

Largest cell – Ostrich egg

Smallest cell – Mycoplasma

Robert Hooke – Micrographia

J.E. Purkinje – Protoplasm

17. Option (4)

Robert Hooke observed honeycomb like structures in cork. he wrote a book micrographia.

18. Option (2)

Anton van Leeuwenhoek was the first one to observe free living cells. the term cell was given by Robert Hooke. cell theory was proposed by Schleiden and Schwann. all cells arise from pre-existing cells was suggested by Virchow.

19. Option (4)

The cork cells as seen by Robert Hooke were published in his book micrographia.

20. Option (4)

The cell theory states that –

Every new cell arises from pre-existing cells.

Bodies of living beings are made up of cells.

Activities of an organism are the sum total of activities of its cells.

Test Subjective Solutions

1. Mitochondria, Plastid
 2. *Amoeba*, *Paramecium*.
 3. Diffusion
 4. (i) Function of leucoplast is storage of oil, starch and proteins.
(ii) Function of chromoplast is pigment synthesis and storage.
 5. Mitochondria are semi-autonomous cell organelles as they have their own DNA and ribosomes which are useful in protein synthesis.
 6. Transport of substances across plasma membrane against the concentration gradient with expenditure of energy is called active transport and transport of substances across plasma membrane along the concentration gradient i.e. from higher concentration to lower concentration without expenditure of energy is called passive transport.
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